Hide menu

Module 5 Introduction

Lesson 1: Parameterized Priors and the Concept of Exchangeability

Lesson 2: Analysis and Applications of Hierarchical Models

Lesson 3: Computational Techniques and Model Validation

Reading: Computational Techniques and Model Validation Reading
2h

Video: Model Validation Pt. 1
4 min

Video: Model Validation Pt. 2
8 min

Practice Assignment: Computational Techniques and Model Validation Quiz
Submitted

Lab: Module 5 - Lesson 3 - RStudio Lab
1h

Module 5 Summative Assessment

Module 5 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Computational Techniques and Model Validation Quiz

Instructions

Review Learning Objectives

1. What is the main advantage of the hierarchical Bayesian approach in the eight schools example? **1 / 1 point**

- It is computationally simpler.
- It ignores variability between schools.
- It is more exchangeable than separate models, capturing the repetition of the experiments at the schools.
- It treats experiments as distinct but related.

Correct
Correct! The hierarchical Bayesian approach provides a compromise between pooling and separate estimates.

Assignment details
Submitted Dec 6, 1:17 AM IST · Attempts Unlimited

Retry

2. Why do we compute pair plots? **1 / 1 point**

- To calculate the means and variances of variables along the diagonal elements.
- To visually inspect for strong linear patterns.
- To perform statistical hypothesis tests.
- To ensure that the data is normally distributed.

Like Dislike Report an issue

Correct
Correct! Pair plots allow us to visually inspect the relationships between all pairs of variables, helping to identify patterns, correlations, divergent transitions and potential outliers.

3. What are divergent transitions? **1 / 1 point**

- Transitions where sampling has errors due to the posterior distribution's geometry.
- Transitions that ensure the sampler has explored the entire parameter space.
- Transitions that show the acceptance rate of the sampler is high and approach infinity asymptotically.
- Transitions that indicate the Markov chain has converged to the target distribution.

Correct
Correct! Divergent transitions occur when the Hamiltonian Monte Carlo (HMC) sampler struggles to accurately follow the geometry of the posterior distribution, often due to regions of high curvature or poor parameter scaling.

Your grade
100%
To pass you need at least 80%. We keep your highest score.
View submission See feedback